USER GUIDE
Windowing Function Figure of Merit Calculator for HP15c CE

(rev1)

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Introduction:

This software pac calculates the figures of merit for several windowing functions used in signal processing. It also allows the user to define their own windowing functions and have these figures calculated.

Figures of Merit Calculated:

1. Coherent Loss
2. Equivalent Noise Bandwidth
3. Processing Loss
4. Scalloping Loss
5. Worst-Case Loss
6. Frequency Response evaluated at frequency f

Window IDs for the built-in the software pac

0. Rectangular (Direclet)
1. Triangular (Fejer / Bartlet)
2. $\cos^{\alpha}(n)$ (where Hann is a special case of these when $\alpha = 2.0$)
3. Hamming
4. IDs 4 – 9 , B – E and .2 – .9 are open for custom windows

Who is this for?

This is for anyone who is ever in dire need of calculating the figures of merit of a built-in or custom-made windowing function.

Usage:

Load Program in format of choice:

dsp_windows

or

dsp_windows_smaller

Prerequisites:

Stack State: N/A

Registers:

`R2` : *Window ID* (see above)

`R3` : Number of points *N*

`R4` : α coefficient

`R5` : Frequency *f* at which to evaluate frequency response where:

f = *(frequency of interest / Sampling Frequency) * N*

Usage:

Press **`GSB A`**

Output:

Figures of Merit in `Matrix E`

Element	Figure of Merit	Units	Units _smaller
`E[1]`	Coherent Gain	linear	linear
`E[2]`	ENBW	bins	bins
`E[3]`	Processing Loss	dB	power
`E[4]`	Scalloping Loss	dB	linear
`E[5]`	Worst-Case Loss	dB	power
`E[6]`	|H(f)|	dB	linear
`E[7]`	arg H(f)	rad	rad

Registers used: `R0`, `R1`, `R2`, `R3`, `R4`, `R5`, `R.1`, `R.3`, `R.4`, `R.5`, `R.6`, `R.7`, `R.8`, `R.9`

Labels used: `A`, `.0`, `.1`, `0`, `1`, `2`, `3`, `D`, `E` (D and E not used in smaller version)

If the user doesn't want the above table in dB and/or prefers saving program memory, then load the *_smaller* version of this program. The software pac will include both versions.

`How to add custom-made windowing functions:`

Go to bottom of program listing and pick a label. This version uses 0-3, so LBL 4 is a good option but it can be any label that's free on the calculator.

Type in the code that generates the custom `w(n)`. Use built-in windows code as reference. The iteration from `0` → `N-1` is done elsewhere, so no need to include it under the custom program. Please note that looping variable *n* is in `R.1`, *N* is in `R3`, and α is in `R4`. Do not modify `R.1` and `R3` in your program.